

PORT-MENIER, QC, Canada

WMO: 718100

Lat: 49.839N

Lon: 64.293W

Elev: 55

StdP: 100.66

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.7	-20.4	-26.9	0.3	-21.9	-24.7	0.4	-20.0	14.9	-5.9	13.3	-7.8	2.7	290	0.722

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.2	23.8	17.9	22.2	17.1	20.9	16.2	19.1	22.1	18.2	20.7	17.3	19.7	4.7	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.1	20.2	17.1	12.3	19.5	16.2	11.6	18.7	54.9	22.1	51.8	20.8	49.0	19.7	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
11.9	10.3	9.1	-27.4	27.3	2.2	1.8	-29.0	28.5	-30.3	29.6	-31.6	30.6	-33.2	31.9		
			-27.6	21.2	2.2	1.1	-29.1	22.0	-30.4	22.7	-31.6	23.3	-33.2	24.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	2.9	-10.6	-10.5	-6.0	0.2	6.4	12.4	16.5	16.0	11.3	5.3	-0.7	-6.5
	DBStd	10.30	5.44	5.51	5.13	3.43	3.09	3.21	2.37	2.42	3.18	3.43	4.22	4.89
	HDD10.0	3141	638	575	497	294	119	13	0	0	21	151	320	512
	HDD18.3	5657	897	808	755	544	371	180	67	79	211	404	570	771
	CDD10.0	543	0	0	0	0	7	84	201	186	61	4	0	0
	CDD18.3	18	0	0	0	0	0	2	9	7	1	0	0	0
	CDH23.3	63	0	0	0	0	0	9	34	18	2	0	0	0
	CDH26.7	3	0	0	0	0	0	1	2	1	0	0	0	0
Wind	WSAvg	4.0	4.2	4.1	4.3	4.3	3.9	3.8	3.7	3.6	3.9	4.0	4.0	4.2
Precipitation	PrecAvg	1220	109	94	89	78	92	93	103	107	104	123	112	116
	PrecMax	1454	156	160	211	139	187	170	162	194	190	222	175	281
	PrecMin	948	40	35	23	26	25	44	54	16	29	41	47	47
	PrecStd	129	33	27	42	32	40	29	31	35	37	40	32	46
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.2	2.9	5.8	11.5	19.8	24.3	26.2	25.5	21.9	15.8	10.5	5.4
		MCWB	2.3	2.3	2.5	5.9	12.2	17.0	19.3	18.9	16.5	12.1	9.9	5.1
	2%	DB	1.2	1.1	3.4	8.4	16.5	21.7	24.2	23.4	19.6	13.5	8.2	3.5
		MCWB	0.8	0.5	1.2	4.4	11.0	15.4	18.1	17.9	15.3	11.0	7.4	3.0
	5%	DB	-0.2	-0.2	2.0	6.5	14.1	19.9	22.6	21.9	17.9	11.9	6.5	1.9
		MCWB	-0.6	-0.8	0.2	3.5	9.0	14.1	17.4	17.1	14.5	10.2	5.7	1.3
	10%	DB	-2.3	-2.1	0.8	4.9	12.1	18.2	21.2	20.6	16.3	10.5	5.0	0.5
		MCWB	-2.8	-2.9	-0.6	2.7	8.1	13.3	16.7	16.4	13.5	8.8	4.2	-0.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.6	2.6	3.3	6.9	14.0	19.0	21.0	20.3	18.1	13.2	10.1	5.1
		MCDB	2.8	2.9	5.0	8.9	17.8	23.2	24.5	23.5	20.0	14.7	10.2	5.3
	2%	WB	0.9	0.6	1.7	5.4	11.3	16.7	19.4	19.0	16.5	11.7	7.7	3.0
		MCDB	1.2	0.9	2.8	7.5	15.0	19.7	22.3	21.9	18.4	12.9	8.1	3.2
	5%	WB	-0.5	-0.8	0.6	4.1	9.9	15.3	18.4	18.1	15.2	10.4	5.8	1.3
		MCDB	-0.2	-0.3	1.6	5.9	13.3	18.6	21.0	20.6	16.9	11.7	6.3	1.7
	10%	WB	-2.7	-2.8	-0.3	2.9	8.5	14.1	17.6	17.2	14.0	9.0	4.3	-0.2
		MCDB	-2.3	-2.2	0.7	4.5	11.5	17.0	20.1	19.5	15.9	10.1	5.0	0.3
Mean Daily Temperature Range	5% DB	MDBR	7.7	7.9	7.9	6.8	8.2	8.6	8.2	8.0	7.6	6.4	5.6	6.3
		MCDBR	8.8	6.6	7.6	9.3	12.3	11.3	10.3	9.7	9.4	8.2	6.4	5.9
		MCWBR	8.9	6.5	6.0	6.2	7.3	6.0	5.3	5.2	6.1	6.4	6.2	6.2
	5% WB	MCDBR	8.9	6.5	7.2	8.0	11.0	9.5	8.4	8.1	8.2	7.3	6.2	6.3
		MCWBR	9.1	6.5	6.3	6.0	7.1	5.7	5.0	5.0	6.0	6.3	6.3	6.5
Clear-Sky Solar Irradiance	taub	0.271	0.276	0.298	0.338	0.357	0.376	0.410	0.377	0.352	0.333	0.295	0.275	
	taud	2.411	2.346	2.377	2.385	2.419	2.388	2.356	2.453	2.483	2.505	2.512	2.413	
	Ebn at Noon	792	876	916	908	901	882	844	860	846	800	759	728	
	Edh at Noon	62	84	97	109	110	115	117	101	89	73	57	54	
All-Sky Solar Radiation	RadAvg	0.98	1.97	3.26	4.45	5.20	5.67	5.40	4.63	3.48	2.12	1.08	0.75	
	RadStd	0.10	0.16	0.28	0.36	0.31	0.49	0.49	0.24	0.24	0.15	0.11	0.08	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (10 neighbors)	N/A	N/A	+1.24	N/A	+0.33	+0.36	N/A	N/A	N/A	N/A

Nomenclature: See separate page